

Recurrent Neural Networks: General Concepts

- ❑ Modeling the time dimension
 - ❑ Adding feedback loops connected to past decisions
 - ❑ Long-term dependencies: Use hidden states to preserve sequential information
- ❑ Training RNNs: Compute a gradient with BPTT (backpropagation through time)
- ❑ Major obstacles of RNN: Long-term dependence
 - ❑ An example: “The Urbana Sweetcorn Festival is held in every August There will be live music, food, beers Enjoy the fresh corn.”
 - ❑ Long-term dependence calls for deep RNN, which makes gradient vanishing and/or exploding problems even worse.
 - ❑ Solutions:
 - ❑ Gated RNNs (LSTM, GRU)
 - ❑ Attention mechanism (Transformer, BERT)

LSTM: Long Short-term Memory

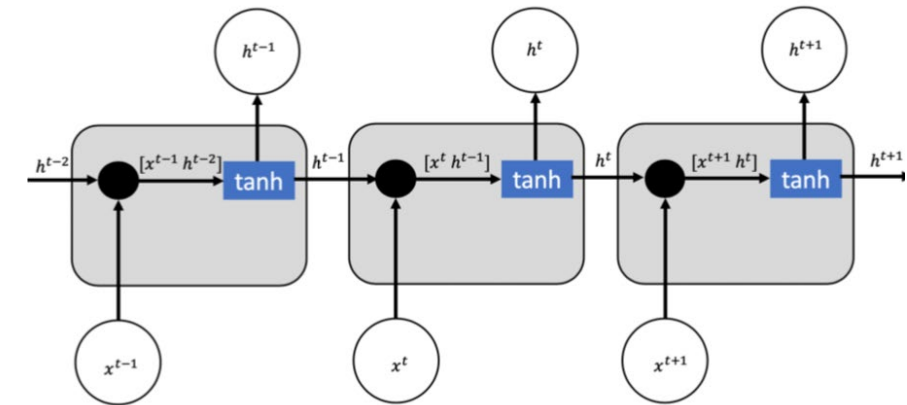
□ RNNs

- Repeatedly apply same activation function
- Rapid decrease of influence

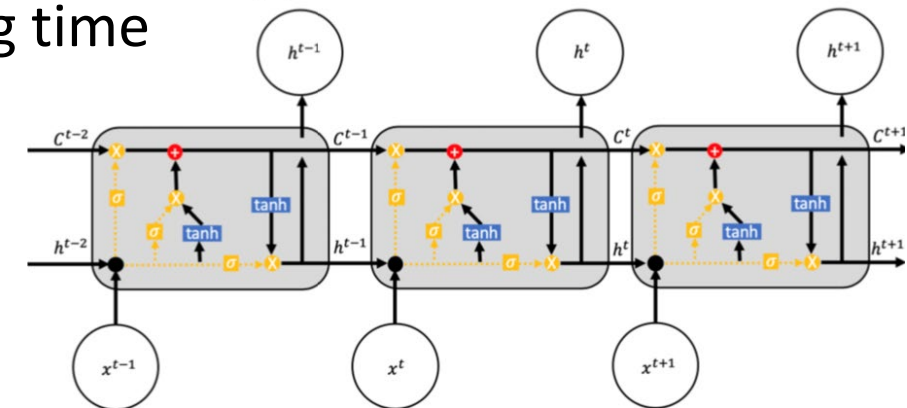
□ LSTM

- Introduce the cell state
- Accumulate the information from the past for a long time
- Three gates to control info flow

- Gray box: RNN cell/LSTM cell
- Black dot: vector concatenation



(a) A traditional RNN model



(b) A long short-term memory (LSTM) model

LSTM Cell

Three Gates: control info flow (blown)

Forget gate $f^t = \sigma(\mathbf{W}_f[x^t, h^{t-1}] + a_f)$

Input gate $i^t = \sigma(\mathbf{W}_i[x^t, h^{t-1}] + a_i)$

Output gate $o^t = \sigma(\mathbf{W}_o[x^t, h^{t-1}] + a_o)$

Temporary Cell State (middle, blue)

$$\tilde{C}^t = \sigma(\mathbf{W}_c[x^t, h^{t-1}] + a_c)$$

Cell state update (upper)

$$C^t = \underbrace{f^t}_{\text{forget gate}} \cdot \underbrace{C^{t-1}}_{\text{old cell state}} + \underbrace{i^t}_{\text{input gate}} \cdot \underbrace{\tilde{C}^t}_{\text{temporary cell state}}$$

Hidden state update (top right)

$$h^t = \underbrace{o^t}_{\text{output gate}} \cdot \underbrace{\tilde{h}^t}_{\text{temporary hidden state}}$$

LSTM variants: GRU

